(SGD), to better scale to large data sets. As the loss in (5) sums over risk sets $\mathcal{R}_i$, which can be as large as the full data set, it cannot be computed in batches. Nevertheless, it is possible to do batched iterations by subsampling the data set (to a batch) and restrict the set $\mathcal{R}_i$ to only contain individuals in the current batch. This scales well for proportional methods such as DeepSurv (Katzman et al., 2018), but would be very computationally expensive for our non-proportional extension presented in Section 3.4. Hence, we propose an approximation of the loss that is easily batched.

Intuitively, we can approximate the risk set $\mathcal{R}_i$ with a sufficiently large subset $\tilde{\mathcal{R}}_i$, and weight the likelihood accordingly with weights w_i,

$$L = \prod_i \left(\frac{\exp[g(\mathbf{x}_i)]}{w_i \sum_{j \in \tilde{\mathcal{R}}_i} \exp[g(\mathbf{x}_j)]} \right)^{D_i}. \tag{7}$$

The weights should ensure that the weighted sum over the subset $\tilde{\mathcal{R}}_i$ in (7) is a reasonable approximation of the full sum over $\mathcal{R}_i$ in (4). By choosing a fixed sample size of the sampled risk sets $\tilde{\mathcal{R}}_i$, we can now optimize the objective by batched gradient descent. The individual i is always included in the sampled risk set $\tilde{\mathcal{R}}_i$ to ensure that each of the products in (7) is bounded above by 1. As the weights w_i do not contribute to the gradients of the logarithm of (7) (as can be seen by differentiating with respect to the model parameters), we can simply drop them from the loss function. Also, in practice we do not compute the loss for $D_i = 0$ as these entries do not contribute to (7). Finally, if we average the loss to make it independent of the data set size, we obtain

$$\text{loss} = \frac{1}{n} \sum_{i:D_i=1} \log \left(\sum_{j \in \tilde{\mathcal{R}}_i} \exp[g(\mathbf{x}_j) - g(\mathbf{x}_i)] \right), \tag{8}$$

where n denotes the number of events in the data set. In our experiments in Sections 5 and 6, we find that it is often sufficient to sample only *one* individual j from the risk set, which gives us the loss

$$\text{loss} = \frac{1}{n} \sum_{i:D_i=1} \log \left(1 + \exp[g(\mathbf{x}_j) - g(\mathbf{x}_i)]\right), \quad j \in \mathcal{R}_i \backslash \{i\}. \tag{9}$$

One benefit of (8) is that it is, in a sense, more interpretable than the negative partial log-likelihood in (5). Due to the sample dependence in the mean partial log-likelihood (MPLL), i.e., the expression in (5) divided by n, the magnitude of the MPLL is dependent on the size of the risk sets. Hence, for a change of batch size, the mean partial log-likelihood changes. This prohibits a comparison of losses across different batch sizes. Comparably, the loss in (8) is not affected by the choice of batch size, as the size of $\tilde{\mathcal{R}}_i$ is fixed. As a result, we can derive the range of values we expect the loss to be in. Using (9) as an example, we know that it is typically in the range (0, 0.693], as a trivial $g(\mathbf{x}) = \text{const}$, gives a loss $= \log(2) \approx 0.693$, and the minimum is obtained by letting $g(\mathbf{x}_i) \to \infty$, $g(\mathbf{x}_j) \to -\infty$, which results in a loss that tends towards 0.

Sampling of the risk sets in Cox's partial likelihood is commonly done in epidemiology and formalized through the nested case-control design, originally suggested by Thomas

(1977). In (8), *case* refers to the i's, while the *controls* are the j's sampled from $\mathcal{R}_i \backslash \{i\}$. Goldstein and Langholz (1992) show that for the Cox partial likelihood, the sampled risk sets produce consistent parameter estimators. While their results do not extend to non-linear models, it is still an indication that the loss function in (8) is reasonable.

Our sampling strategy deviates from that of the nested case-control literature in two ways. Firstly, we sample a new set of controls for every iteration, instead of keeping control samples fixed. Secondly, we sample controls with replacement, as this requires less computation than sampling without replacement. Note, however, that we typically sample a single control, in which case it does not matter if we sample with or without replacement.

3.3 Non-Linear Cox

Having established the simple loss function in (8), which can be computed with SGD, the generalization of the relative risk function $\exp[g(\mathbf{x})]$ is rather straightforward. In this paper, we replace the linear predictor $g(\mathbf{x}) = \boldsymbol{\beta}^T \mathbf{x}$ by a $g(\mathbf{x})$ parameterized by a neural network. While our proposed loss function is not a requirement for the adaptation of a neural network (see, e.g., DeepSurv by Katzman et al., 2018), it really helps for the further extensions in Section 3.4. Also, it has been found that batched iterations can improve predictive performance (Keskar et al., 2016; Hoffer et al., 2017).

Our generalization of $g(\mathbf{x})$ leaves the presented theory in Sections 3.1 and 3.2 essentially unchanged, so we do not repeat the likelihoods and loss functions for this model.

We will later refer to the Cox proportional hazards model parameterized with a neural network as Cox-MLP. To differentiate between minimizing the negative partial log-likelihood in (5), as done by DeepSurv, and our case-control approximation in (8), we will denote the corresponding methods by Cox-MLP (DeepSurv) and Cox-MLP (CC), respectively.

For the non-linear Cox models, the loss does not necessarily have a unique minimizer for $g(\mathbf{x})$. Therefore, we add a penalty to the loss function to encourage $g(\mathbf{x})$ to not deviate too far from zero

$$\text{penalty} = \lambda \sum_{i:D_i=1} \sum_{j \in \tilde{\mathcal{R}}_i} |g(\mathbf{x}_j)|. \tag{10}$$

Here λ is a tuning parameter, and note that i is included in $\tilde{\mathcal{R}}_i$.

3.4 Non-Proportional Cox-Time

The proportionality assumption of the Cox model can be rather restrictive, and parameterizing the relative risk function with a neural net does not affect this constraint. Approaches for circumventing this restriction are typically based on grouping the data based on a categorical covariate and applying a stratified version of the Cox model (Klein and Moeschberger, 2003, chap. 9). We propose a parametric approach that does not require stratification. Continuing with the semi-parametric form of the Cox model, we now let the relative risk function depend on time,

$$h(t \,|\, \mathbf{x}) = h_0(t) \exp[g(t, \mathbf{x})]. \tag{11}$$

In practice, we let $g(t, \mathbf{x})$ handle the time as a regular covariate, which enables $g(t, \mathbf{x})$ to model interactions between time and the other covariates. This is similar to the approach taken in classical survival analysis, where the non-proportional effect of a covariate x may be modeled by including time-dependent covariates like $x \cdot t$ and $x \cdot \log t$.

The model (11) is no longer a proportional hazards model. However, it is still a relative risk model with the same partial likelihood as previously, only now with an additional covariate. Following the approach from Section 3.2, we have the loss function

$$\text{loss} = \frac{1}{n} \sum_{i:D_i=1} \log \left(\sum_{j \in \tilde{\mathcal{R}}_i} \exp[g(T_i, \mathbf{x}_j) - g(T_i, \mathbf{x}_i)] \right), \tag{12}$$

and we include the penalty in (10), with $g(T_i, \mathbf{x}_j)$ replacing $g(\mathbf{x}_j)$. We will later refer to models fitted by (12) as *Cox-Time*.

Note that the loss has the same T_i for both $\mathbf{x}_i$ and the $\mathbf{x}_j$'s. Consequently, if we had used the full risk set $\mathcal{R}_i$ instead of the subset $\tilde{\mathcal{R}}_i$, as is the case for the loss in (5), the loss would become very computationally expensive. In fact, for the full risk set, the time complexity of the loss would be $O(n \cdot |\mathcal{R}_i|) = O(n^2)$, where $|\mathcal{R}_i|$ denotes the size of the risk set. But for (12) we get $O(n \cdot |\tilde{\mathcal{R}}_i|) = O(n)$, as $|\tilde{\mathcal{R}}_i|$ is fixed and small. In the proportional case, to compute the loss in (5) one only needs to compute $g(\mathbf{x}_j)$ once (per iteration) for each j, and reuse that value in all other risk sets. This ensures the linear time complexity for the classical Cox models.

We can find the Breslow estimate for the cumulative baseline hazard $H_0(t)$ using (6) with $\hat{g}(\mathbf{x}_j)$ replaced by $\hat{g}(T_i, \mathbf{x}_j)$. Note that we still need the non-parametric baseline, as $g(t, \mathbf{x})$ is restricted to model interactions between the time and the covariates. To see this, consider $g(t, \mathbf{x}) = a(t, \mathbf{x}) + b(t)$, and observe that $b(t)$ cancels out in the loss.

3.5 Prediction

We can obtain predictions from the relative risk models by estimating the survival function in (1), $\hat{S}(t \,|\, \mathbf{x}) = \exp[-\hat{H}(t \,|\, \mathbf{x})]$. For the proportional hazards models, the relative risk function does not depend on time, enabling us to integrate only over the baseline hazard and compute the relative risk separately,

$$H(t \,|\, \mathbf{x}) = \int_0^t h_0(s) \exp[g(\mathbf{x})] \, ds = H_0(t) \exp[g(\mathbf{x})].$$

By first estimating $H_0(t)$ on the training data with (6), we only need to compute $\exp[g(\mathbf{x})]$ to obtain predictions. Computation of the estimate $\hat{H}_0(t)$ requires a single pass over the whole training set, in addition to sorting the training set by time.

In the case of models with non-proportional hazards, such as models fitted by Cox-Time in Section 3.4, predictions are much more computationally expensive. As the relative risk is time-dependent, we now need to integrate over both the baseline hazard and $g(t, \mathbf{x})$,

$$H(t \,|\, \mathbf{x}) = \int_0^t h_0(s) \exp[g(s, \mathbf{x})] \, ds.$$

In practice, we estimate the cumulative hazards by

$$\hat{H}(t \mid \mathbf{x}) = \sum_{T_i \leq t} \Delta \hat{H}_0(T_i) \exp[\hat{g}(T_i, \mathbf{x})],$$

where $\Delta \hat{H}_0(T_i)$ is an increment of the Breslow estimate and $\hat{g}(T_i, \mathbf{x})$ is the estimate of $g(T_i, \mathbf{x})$ obtained from the neural network. This is clearly rather computationally expensive as we need to compute $\hat{g}(T_i, \mathbf{x})$ for all distinct event times $T_i \leq t$. Furthermore, for continuous-time data, computation of the cumulative baseline hazard through the Breslow estimate,

$$\Delta \hat{H}_0(T_i) = \frac{D_i}{\sum_{j \in \mathcal{R}_i} \exp[\hat{g}(T_i, \mathbf{x}_j)]}, \tag{13}$$

scales quadratically.

To alleviate the computational cost, one can compute the cumulative hazards over a reduced number of distinct time points. Hence, Cox-Time is trained on continuous-time data but produces discrete-time predictions, with the benefit of the discretization happening after the network is fitted. In practice, we perform this discretization by computing the baseline on a random subset of the training data and subsequently control the resolution of the time grid through the sample size.

4. Evaluation Criteria

Metrics for evaluating the performance of methods for time-to-event prediction should account for the censored individuals. In the following, we describe the metrics used in the experimental sections of this paper.

4.1 Concordance Index

In survival analysis, the concordance index, or C-index (Harrell Jr et al., 1982), is arguably one of the most commonly applied discriminative evaluation metrics. This is likely a result of its interpretability, as it has a close relationship to classification accuracy (Ishwaran et al., 2008) and ROC AUC (Heagerty and Zheng, 2005). In short, the C-index estimates the probability that, for a random pair of individuals, the predicted survival times of the two individuals have the same ordering as their true survival times. See Ishwaran et al. (2008) for a detailed description.

As the C-index only depends on the ordering of the predictions, it is very useful for evaluating proportional hazards models. This is because the ordering of proportional hazards models does not change over time, which enables us to use the relative risk function instead of a metric for predicted survival time. It is, however, not obvious how the C-index should be applied for non-proportional hazards models (Gerds et al., 2012; Ishwaran et al., 2008). We will use a metric based on the time-dependent C-index by Antolini et al. (2005), which estimates the probability that observations i and j are concordant given that they are comparable,

$$C^{\text{td}} = \text{P}\{\hat{S}(T_i \mid \mathbf{x}_i) < \hat{S}(T_i \mid \mathbf{x}_j) \mid T_i < T_j, D_i = 1\}. \tag{14}$$